

3(a)	<u>sum / total</u> momentum before (a collision) = <u>sum / total</u> momentum after (a collision) or <u>sum / total</u> momentum (of a system) is constant	M1
	if no (resultant) external force (acts) / for an isolated system	A1
3(b)	along direction of motion: $10m = 2mv \cos 30^\circ + 3mw \cos 30^\circ$	C1
	perpendicular to direction of motion: $2mv \cos 60^\circ = 3mw \cos 60^\circ$ $(v = 3w / 2)$	C1
	$v = 2.9 \text{ m s}^{-1}$	A1
	$w = 1.9 \text{ m s}^{-1}$	A1
3(c)	$E_K = \frac{1}{2} \times m \times v^2$ $(= \frac{1}{2} \times 4.2 \times 6.0^2)$ $(= 76 \text{ J})$	C1
	force = work done / distance	C1
	force = $76 / 0.050$ $= 1500 \text{ N}$	A1

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Question	Answer	Marks
3(c)	or	
	$a = (-)u^2 / 2s$ $= (-)6.0^2 / (2 \times 0.050)$ $(= (-)360 \text{ m s}^{-2})$	(C1)
	$F = ma$	(C1)
	$F = 4.2 \times 360$ $= 1500 \text{ N}$	(A1)
	or	
	$a = (-)u^2 / 2s$ $= (-)6.0^2 / (2 \times 0.050)$ $(= (-)360 \text{ m s}^{-2})$	(C1)
	$t = -u / a$ $= -6.0 / -360 (= 0.017 \text{ s})$ $F = \Delta p / t$	(C1)
	$F = (0 - 4.2 \times 6) / 0.017$ $= 1500 \text{ N}$	(A1)